Famicom JAMMA board

power

File: power.kicad_sch

Power: +5V entry, switched +5V_SW, +3V3, supply of the RP2A03 modules

cpu

File: cpu.kicad_sch

CPU: RP2A03, work RAM, address decoder, data-bus resistors

ppu

File: ppu.kicad_sch

PPU: RP2C02, address latch, video RAM, sprite detection

cart

File: cart.kicad_sch

Cartridge connector and the dump circuit for emulation mode

video

File: video.kicad_sch

Video: RGB resistor DAC and composite sync

audio

File: audio.kicad_sch

Audio: mixer, loop through the cartridge, power amplifier

jamma

File: jamma.kicad_sch

JAMMA edge connector, inputs, coin counter and lock-out drivers

mcu

File: mcu.kicad_sch

Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

PWR_FLAG

GND

PWR_FLAG

+5V

PWR_FLAG

VSYS

Sheet: /
File: fcjamma.kicad_sch

Title: Famicom JAMMA board

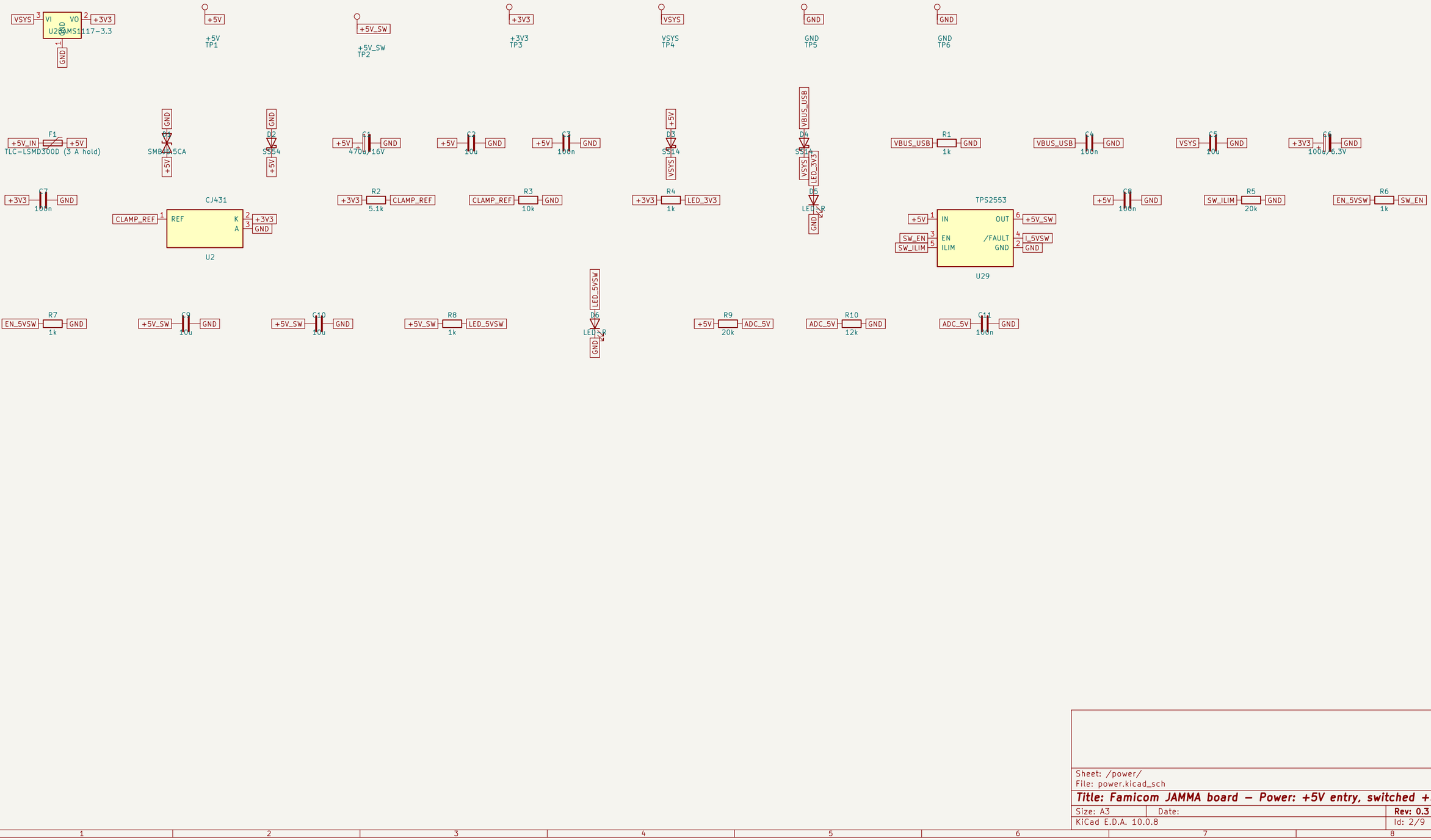
Size: A3	Date:	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 1/9

Power: +5V entry, switched +5V_SW, +3V3, supply of the Pico modules

+5V_IN comes from the JAMMA fingers 3, 4, C, D. −5 V and +12 V are not used. Supply range at the fingers: 4.75 V to 5.25 V.

+5V_SW feeds the CPU, the PPU, the memories, the 5 V logic and the cartridge through the power switch U29. It is off until Pico C raises EN_5VSW, so that no 5 V signal reaches a Pico pin before that Pico is powered (the RP2350 pins tolerate 5 V only while powered). U29 limits the current to about 1.3 A: a short circuit in a cartridge cannot draw the cabinet supply's full current.

VSYS feeds the three Pico modules and the +3V3 regulator from +5V or from the USB socket, whichever is present.



Sheet: /power/
File: power.kicad_sch

Title: Famicom JAMMA board — Power: +5V entry, switched +5V_SW, +3V3, s

Size: A3	Date:	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 2/9

CPU: RP2A03, work RAM, address decoder, data-bus resistors

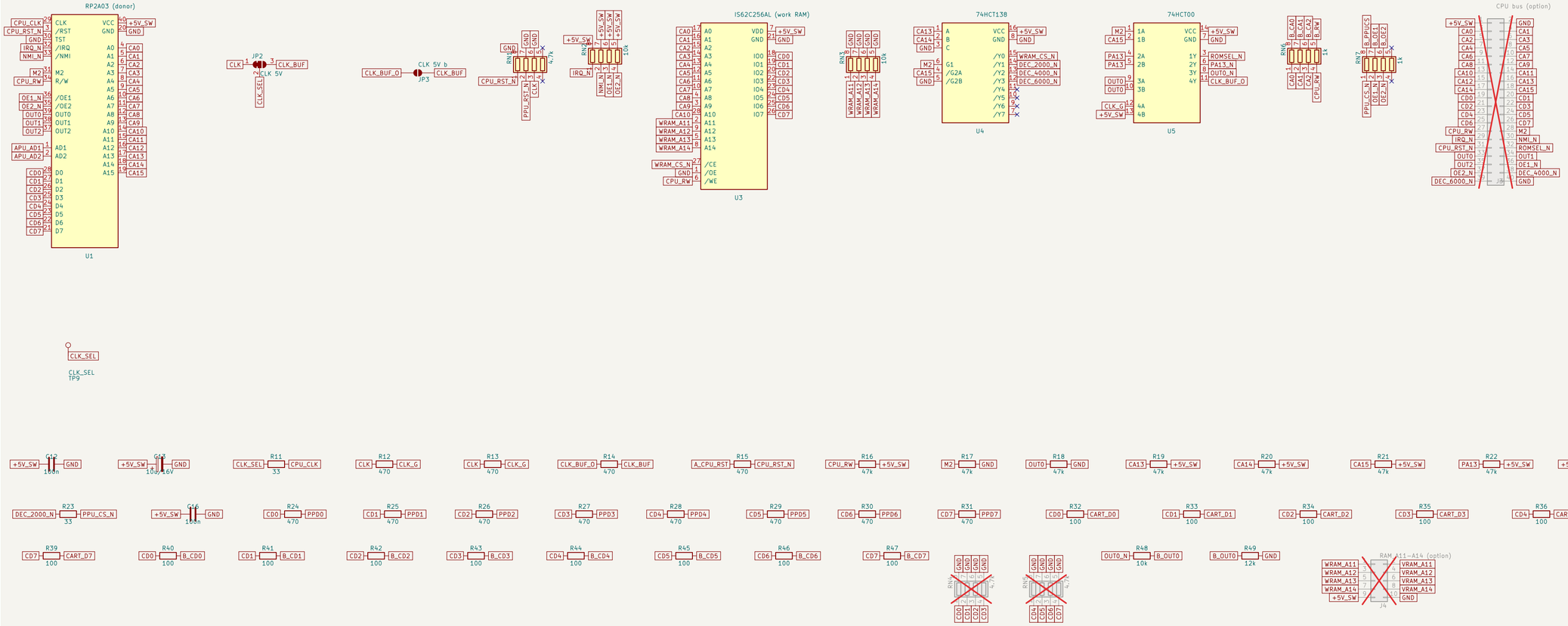
U1 is the donor's RP2A03, plugged into a 40-pin socket. In emulation mode the socket is empty and the dump shift registers drive the buses.

The 470 ohm resistors let Pico B replace what the PPU sees on a write. A line that Pico B pulls low also drags the CPU side down,

so the firmware reads the CPU side first and drives afterwards (README 3.9).

No pull-ups on the CPU data bus: they are known to upset some PPU revisions, and games read the floating bus.

The logic next to the CPU and the PPU is 74HCT: its inputs switch at TTL levels, which is what the two NMOS chips deliver.



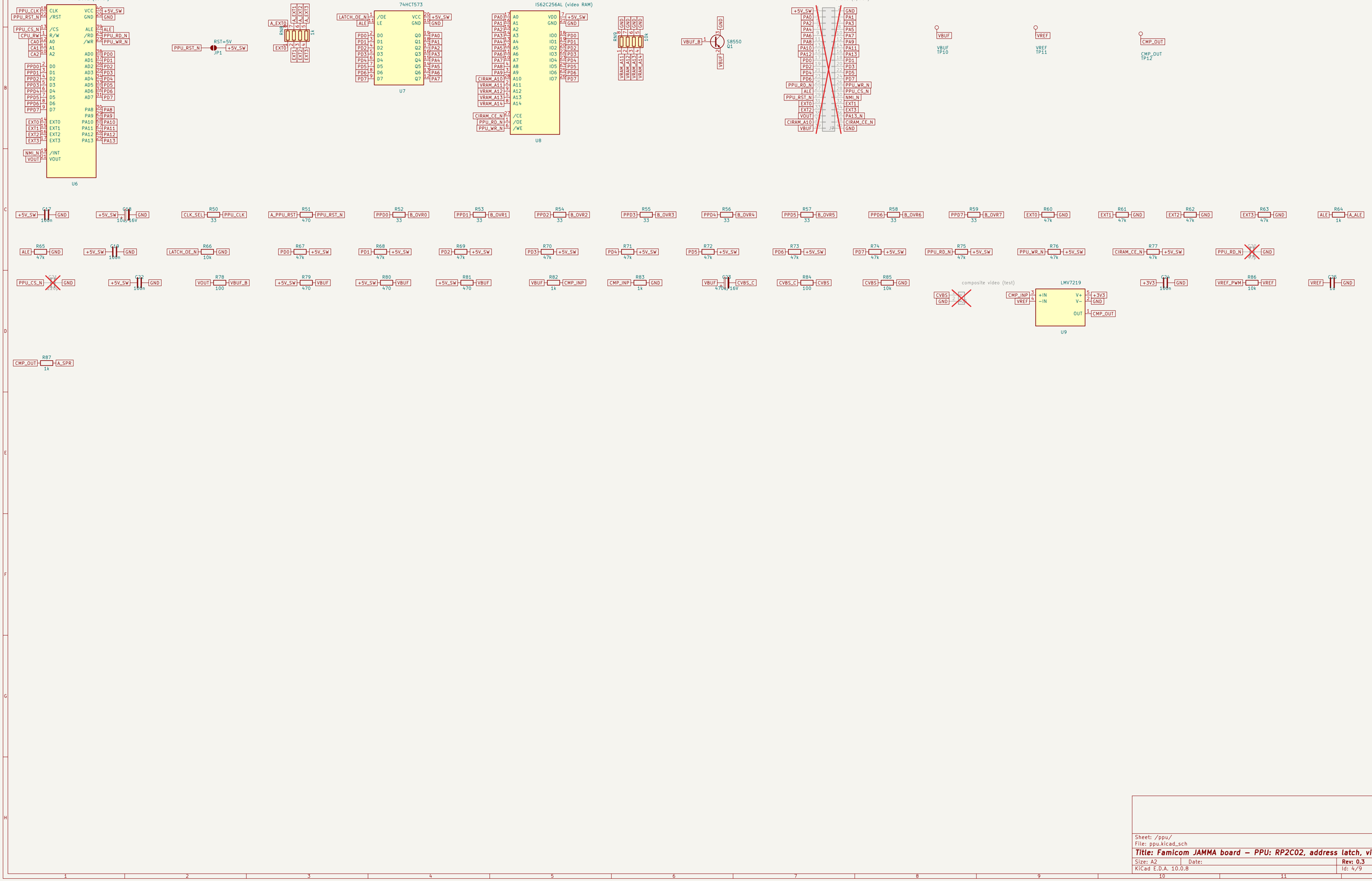
PPU: RP2C02, address latch, video RAM, sprite detector

U6 is the donor's RP2C02, plugged into a 40-pin socket. EXT0–3 give the lower four bits of the palette index of every pixel once

Pico B has forced bit 6 of register 2000 high. The fifth bit (sprite or background) is read from the brightness of VOUT:

Pico B makes the PPU store a dark colour in every background entry and a bright one in every sprite entry.

EXT shows entry 0 of the current background palette (4, 8 or 12) on transparent pixels, and 0 while rendering is off: the firmware maps these.



Cartridge connector and the dump circuit for emulation mode

J8 is a new 60-pin card-edge connector. Pin 45 carries the board's sound to the cartridge, pin 46 brings it back.

Dump circuit: with the CPU and PPU sockets empty, Pico C reads the cartridge through shift registers and hands the data to Pico A.

The six 74HCT595 run from +5V_SW and take the 3.3 V signals of Pico C through 470 ohm.

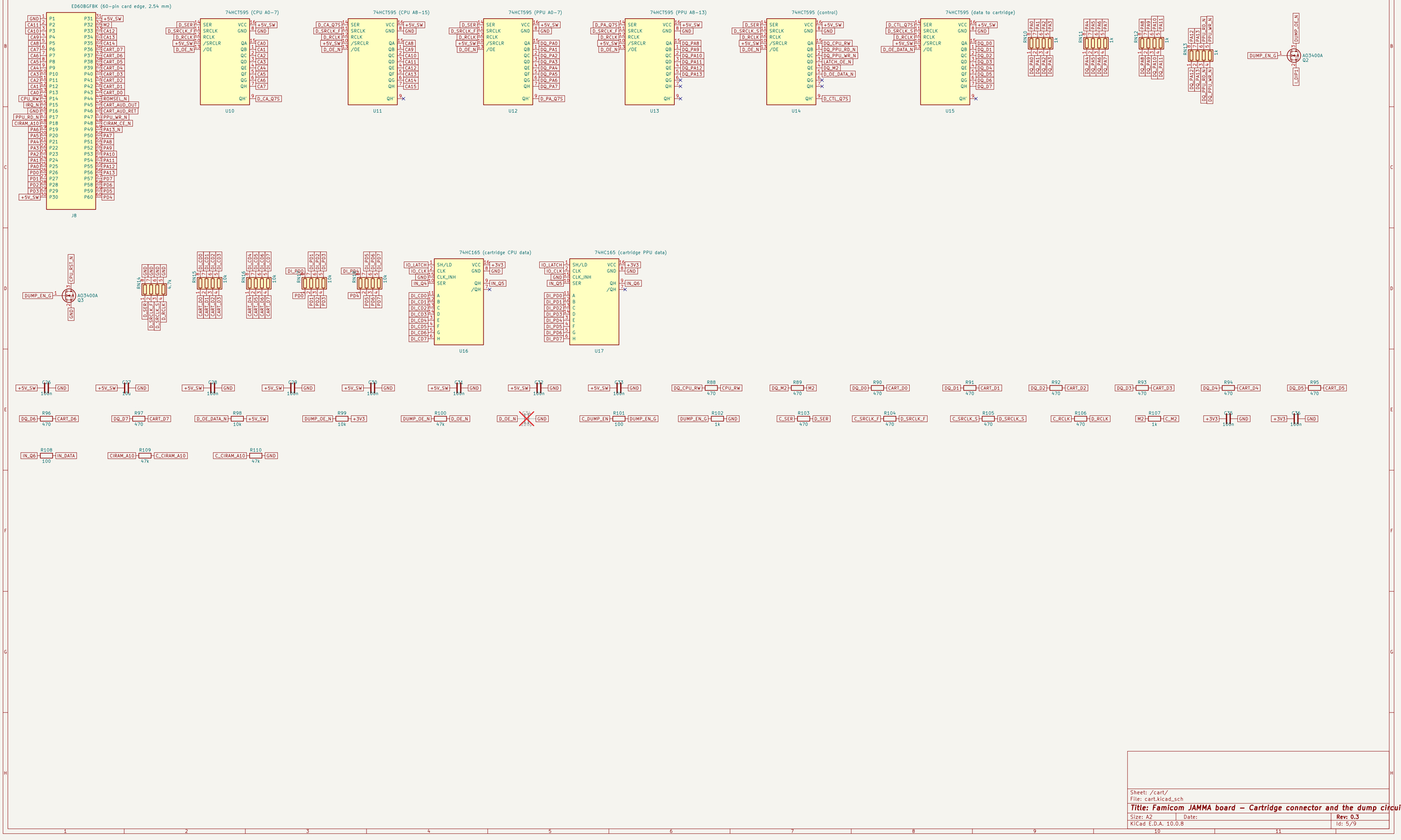
Their outputs can be switched on only while DIP switch 1 is set to emulation AND Pico C raises C_DUMP_EN (Q with its source on the switch).

C_DUMP_EN also holds the CPU in reset (all its bus pins are released then). A PPU cannot be switched off its bus, so every driver that

meets a PPU output has 1 k in series: the drivers put at most 5 mA into each pin of a PPU left in its socket by mistake.

The resistors do not cover PD0–7: a forced /RD sets the cartridge ROM against such a PPU. Only the firmware's check for chips prevents that.

Pico C must hold its four lines to the 74HCT595 low whenever +5V_SW is off.

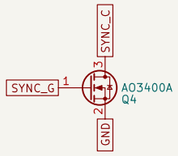


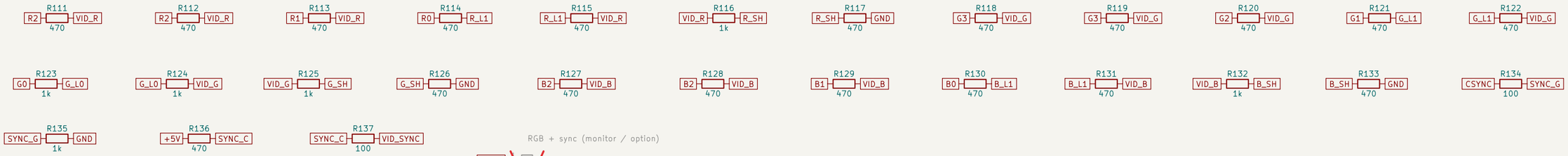
Sheet: /cart/	
File: cart.kicad_sch	
Title: Famicom JAMMA board – Cartridge connector and the dump circuit for	
Size: A2	Date:
KiCad E.D.A. 10.0.8	**Rev: 0.3**
	Id: 5/9

Video: RGB resistor DAC and composite sync

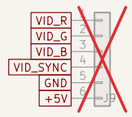
Pico A drives the DAC directly. Full scale is 3.0 V open circuit, about 2.85 V into 2 k, source resistance about 125 ohm.

Red and blue have 3 bits, green has 4. Only 470 ohm and 1 k are used so that the parts list stays short.





RGB + sync (monitor / option)



Sheet: /video/
File: video.kicad_sch

Title: Famicom JAMMA board – Video: RGB resistor DAC and composite sync

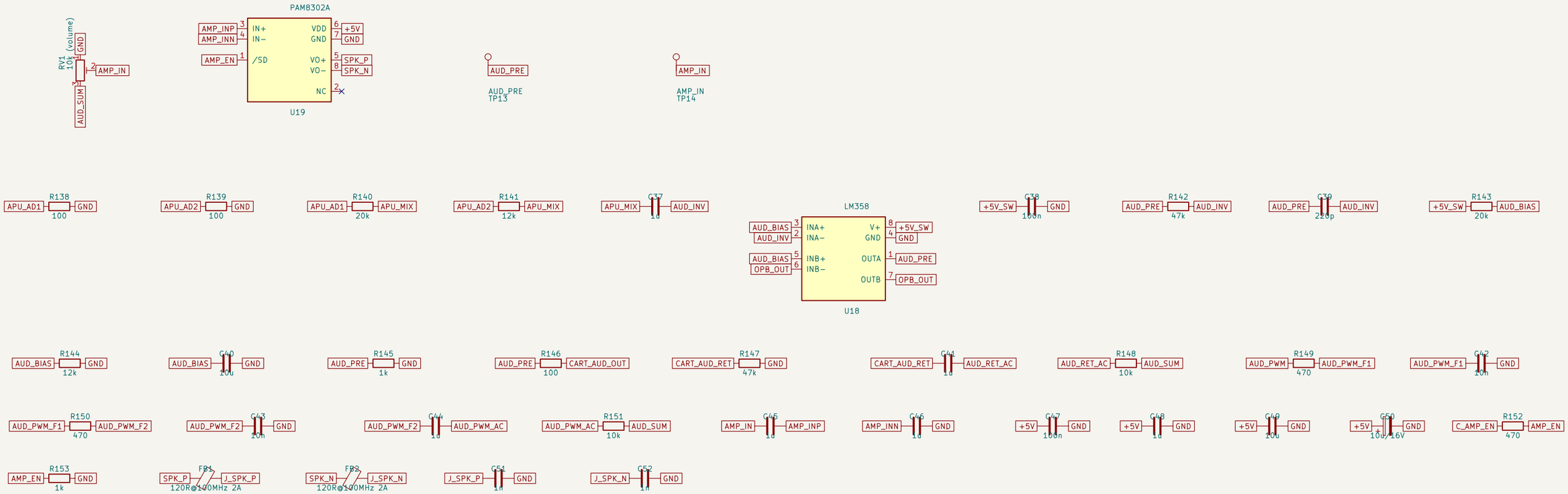
Size: A3	Date:	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 6/9

Audio: mixer, loop through the cartridge, power amplifier

Real-chip mode: 2A03 pins -> mixer (Famicom values) -> cartridge pin 45 -> cartridge -> pin 46 -> volume -> amplifier.

Emulation mode: Pico A's PWM output -> two RC stages -> volume -> amplifier.

The loudspeaker outputs are bridge-tied: JAMMA L is not ground. The amplifier stays off until Pico C raises AMP_EN.



Sheet: /audio/		
File: audio.kicad_sch		
Title: Famicom JAMMA board – Audio: mixer, loop through the cartridge, pow		
Size: A3	Date:	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 7/9

JAMMA edge connector, inputs, coin counter and lock-out drivers

Every input: 1 k pull-up to +3V3 (3.3 mA through the switch), 4.7 k in series, 10 nF to ground, then a 74HC165 on +3V3.

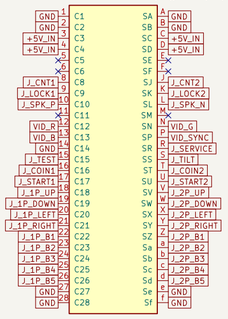
Pico C reads the chain: after a low pulse on IO_LATCH the first bit on IN_DATA is input H of U17, the last is input A of U20.

The same pulse copies the 74HC595's shift register to its outputs, so one cycle reads the inputs and writes the outputs.

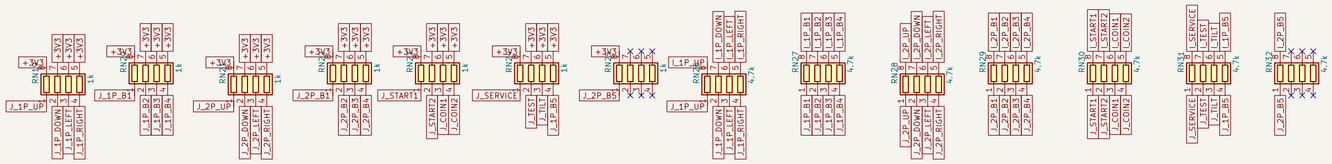
Fingers 5/E (−5 V), 6/F (+12 V) and 11/M are not connected. 7/H is the key slot.

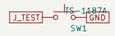
Coin counter and lock-out coils: 5 V or 12 V types, switched to ground; each output has its own clamp, so the board needs no +12 V.

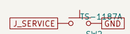
JAMMA edge (gold fingers)

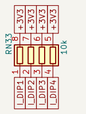


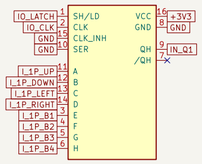
J1



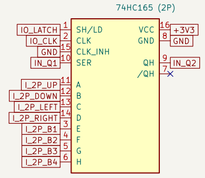




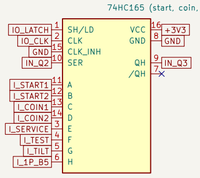




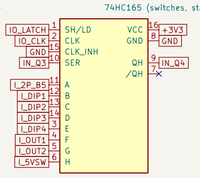
74HC165 (2P)



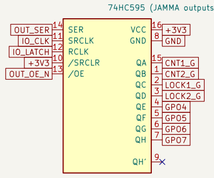
74HC165 (start, coin, service)

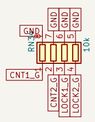


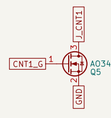
74HC165 (switches, status)

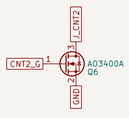


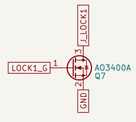
74HC595 (JAMMA outputs)

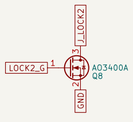


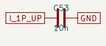


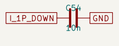






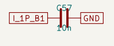


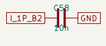


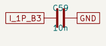




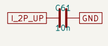




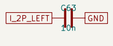




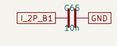


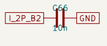












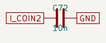




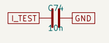








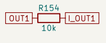




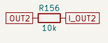




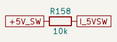
























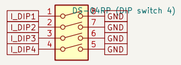














Sheet: /jamma/
File: jamma.kicad_sch

Title: Famicom JAMMA board — JAMMA edge connector, inputs, coin counter a

Size: A2	Date:	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 8/9

Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

Pico A (video): clock and reset for CPU and PPU, palette index and sprite bit in, RGB and sync out; the emulator in emulation mode.

Pico B (bus): watches the CPU's accesses to the PPU, replaces palette data on the PPU side, answers the controller reads.

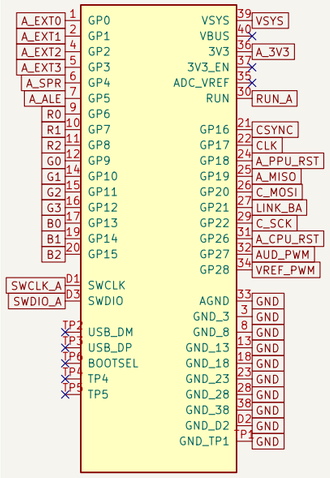
Pico C (input/output and management): JAMMA inputs and outputs, coin and timer logic, cartridge dump, USB Type-C,

power switch of the Famicom half, and reprogramming of Pico A and Pico B through their debug pads.

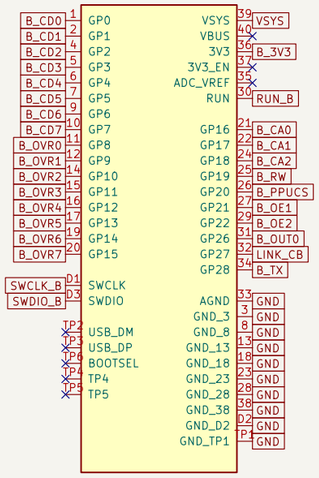
The modules are soldered flat on the board. The USB socket reaches Pico C through the pads under the module (TP2, TP3).

Use either the board's Type-C socket or Pico C's own Micro-USB socket, never both at once: they are the same two wires.

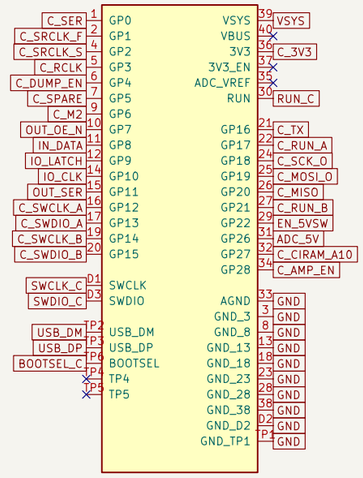
Pico 2 A (video)



Pico 2 B (bus)



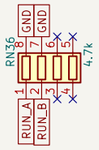
Pico 2 C (I/O, management)







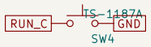


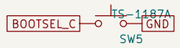




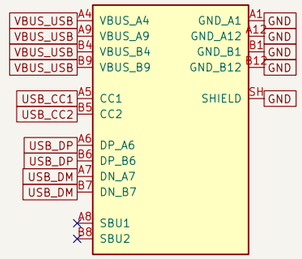


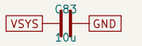






USB Type-C socket (TYPE-C-31-M-12)



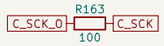




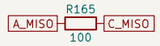








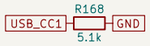


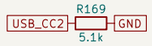














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Title: Famicom JAMMA board – Three Raspberry Pi Pico 2 modules, USB Type

Size: A3 | Date: | **Rev: 0.3**

KiCad E.D.A. 10.0.8 | Id: 9/9